Audio-Based Deepfake Detection Leveraging ECAPA-TDNN and TIMIT-TTS Corpus

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***Abstract*—The improvement of generative speech technologies has made synthetic voices seem more humanlike, resulting in threats such as impersonation, misinformation, and compromised voice identity. We present an audio-only deepfake detector that utilizes ECAPA-TDNN speaker embeddings with a simple binary classifier to mitigate these threats. Our experiments utilize the TIMIT-TTS dataset that has nearly 80,000 samples from twelve text-to-speech systems. The results of our study showed that this embedding-based method performed particularly well, achieving 90% accuracy, F1 of 0.92, AUC of 0.93, and is shown to outperform the RawNet2 baseline consistently across different testing conditions. These results suggest that modeling at the embedding level appears to be a useful compromise between accuracy and efficiency and lends itself readily to real-world applications requiring security.**

***Index Terms*—Audio Deepfake Detection, ECAPA-TDNN, Speaker Embeddings, TIMIT-TTS Dataset, Voice Spoofing, Synthetic Speech Forensics**

I. INTRODUCTION

The rapid evolution of deep learning and generative speech technologies has made it possible to produce synthetic voices that are almost indistinguishable from real human speech. These so-called audio deepfakes can replicate not only a person's tone and accent but also their rhythm and speaking style, making them increasingly convincing. [1] Although such technologies enable useful applications—such as improving accessibility, powering virtual assistants, and generating realistic voiceovers—they also pose serious risks. Malicious actors can exploit synthetic speech for impersonation, spreading misinformation, and bypassing voice authentication systems, raising concerns about privacy, security, and digital trust.

To address these challenges, this work introduces a detection framework that leverages ECAPA-TDNN speaker embeddings as the foundation for distinguishing between real and synthetic speech. [2]Unlike traditional waveform-based detectors, the proposed approach operates in the embedding space, which reduces complexity while retaining strong discriminative power. This design makes the system lightweight, interpretable, and well-suited for real-time deployment. [3] Furthermore, the framework is evaluated on the TIMIT-TTS corpus, a large and diverse dataset containing speech generated by multiple text-to-speech models under realistic post-processing conditions.

The key novelty of this study lies in combining the robustness of ECAPA-TDNN embeddings with a simple yet effective binary classifier, enabling consistent detection performance across varied conditions. [4] By demonstrating superior results compared to RawNet2 and other baseline systems, [5], [6] this work contributes to building reliable and practical defenses against audio deepfakes for applications in voice security, digital media forensics, and trustworthy AI.

The remainder of this paper is organized as follows. Section II reviews earlier research and important background concepts. Section III describes the proposed methodology and system construction. Section IV presents the experimental setup and results. Section V discusses the limitations encountered and possible improvements. Finally, Section VI concludes with the key findings of this work.

II. Related Work

Research on detecting synthetic speech has accelerated in recent years, driven by the rapid realism of modern TTS and voice conversion systems. [7] Broadly, prior work spans three threads: audio-only detectors, multimodal audio-visual models, and methods focused on robustness and adversarial resilience.

Audio-only detectors. A large body of work uses time–frequency representations (e.g., log-Mel, CQCC, CQT) with convolutional or transformer backbones to capture subtle spectro-temporal artifacts introduced by synthesis. These models have steadily improved on standard benchmarks and now report strong AUC/F1 under clean conditions. [8] Parallel efforts explore waveform-level architectures that learn directly from raw audio; while effective with sufficient data, such end-to-end models can be heavy, slower to train, and sensitive to domain shift. [9] Recent studies have therefore emphasized lighter front ends and better inductive bias to improve generalization.

Embedding-based approaches. Another line leverages pretrained speaker-recognition encoders (e.g., x-vectors, ECAPA-TDNN) to extract compact, speaker-discriminative embeddings that also expose synthesis artifacts. [10] These representations enable simple downstream classifiers while remaining efficient and interpretable.

Robustness and generalization. Practical deployments face codec compression, background noise, reverberation, pitch/time modifications, and unseen TTS models. [11]Recent work evaluates cross-corpus and open-set generalization, showing that detectors frequently degrade when the synthesis method or channel conditions change. [12]–[14] Techniques such as augmentation, confidence calibration, domain adaptation, and open-set training have been proposed to narrow [15] this gap, but consistent robustness across distortions remains challenging.

Adversarial resilience. Studies from 2022 onward also examine adversarial examples and intentionally obfuscated audio designed to evade detectors. [16] Defenses include adversarial training, ensemble decision rules, and preprocessing that reduces imperceptible perturbations. [17]These methods help, but often raise computational cost and can introduce trade-offs with clean-speech accuracy.

Research gap. Despite progress, there is still a need for audio-only detectors that (i) remain lightweight and interpretable, (ii) generalize across speakers, synthesis methods, and common post-processing, and (iii) are suitable for real-time use. [18] Embedding-based pipelines built on robust speaker encoders are a promising direction: they retain discriminative power while simplifying the classifier and reducing latency. [19]

III. Proposed Methodology

The proposed framework for audio deepfake detection combines ECAPA-TDNN speaker embeddings with a lightweight binary classifier. This section outlines the experimental setup, dataset, preprocessing steps, embedding extraction, and classification approach.

A. Experimental Setup

All experiments were conducted in a Google Colaboratory Pro+ environment, configured with Python 3.10, PyTorch, and supporting audio libraries such as torchaudio and scikit-learn. The cloud hardware included two virtual CPUs, 12 GB RAM, and an NVIDIA Tesla T4 GPU (16 GB VRAM). This configuration enabled efficient training and evaluation without memory constraints.

The model was trained for 50 epochs with early stopping applied to prevent overfitting, based on validation loss. Optimization was carried out using the Adam algorithm with a learning rate of 1×10^{-3} and a batch size of 32. On average, each experiment required about 2.5 hours, including both embedding extraction and classifier training. To ensure reproducibility, dataset storage and intermediate checkpoints were maintained on Google Drive.

B. Dataset Description

This study uses the TIMIT-TTS corpus [20], a synthetic extension of the original TIMIT dataset. It contains high-quality speech generated by 12 different Text-to-Speech (TTS) models, with each original utterance re-synthesized to create paired real and fake samples. This setup provides a large multi-speaker corpus that supports fair benchmarking of deepfake detection methods under consistent linguistic and acoustic conditions

The dataset comprises nearly 80,000 audio clips from 37 speakers, including both real and synthetic recordings with variations such as compression, pitch shifts, and background noise. All experiments used the publicly available TIMIT-TTS corpus, which contains only synthetic, non-identifiable data. No personal information was involved, ensuring reproducibility while adhering to ethical standards in data handling

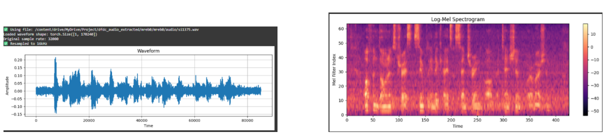
TABLE I: DATASET STATISTICS (TIMIT-TTS SUBSET USED)

Attribute	Value
Total Samples	79,776
Number of Speakers	37
Real Samples	6,648
Fake Samples (TTS)	73,128
TTS Models Used	12
Sampling Rate	16 kHz
Audio Duration	~3 sec avg.
Train/Test Split	80% / 20%

C. Data Preprocessing

Audio inputs were first converted to mono channel and resampled to 16 kHz using torchaudio.transforms.Resample. This ensured consistency with ECAPA-TDNN input requirements. Each waveform was normalized via z-score standardization to stabilize learning and prevent features with larger magnitudes from dominating classification. Labels were mapped to binary values (0 = real, 1 = fake).

Fig. 1 shows an example raw audio waveform, while Fig. 2 presents its corresponding log-Mel spectrogram after preprocessing. These representations illustrate the transformation pipeline prior to embedding extraction.



(a) Raw Audio Waveform (b) Log-Mel Spectrogram

Fig. 1: Preprocessing pipeline: (a) raw audio waveform and (b) its corresponding log-Mel spectrogram.

As shown in Fig. 1, raw audio is first converted into a log-Mel spectrogram, forming the input representation.

To meet the input requirements of the ECAPA-TDNN speaker embedding architecture , all audio samples are resampled to a fixed sampling rate of 16 kHz using `torchaudio.transforms.Resample`. This uniform resampling preserves temporal and spectral characteristics that are essential for effective embedding generation. Preprocessing at the waveform level, including channel handling and resampling, plays a vital role in reducing inconsistencies that may affect downstream speaker representation or classification tasks . Z-score standardization is used to normalize all feature vectors following embedding extraction. `StandardScaler` from the `scikit-learn` library is used to implement this. According to the training distribution, normalization entails scaling the data to unit variance and centering it at zero mean. To avoid features with larger numerical scales dominating statistical classifiers like logistic regression, this technique is especially crucial [21]. Furthermore, to facilitate supervised training and assessment, all string-based class labels—such as ”Real” and ”Fake”—are converted into binary integer labels.

D. ECAPA-TDNN Speaker Embedding Network

An ECAPA-TDNN (Emphasized Channel Attention, Propagation and Aggregation Time Delay Neural Network) convolutional neural architecture was developed to perform reliable speaker embedding extraction out of audio segments. The ECAPA-TDNN advances the original Time-Delay Neural Network (TDNN) by including Res2Net modules and squeeze-and-excitation (SE) blocks well-structured, and incorporates attentive statistical pooling to better extract speaker specific feature across time. To derive contextual representations from an incoming audio waveform $x(t)$ sampled at 16 kHz, the ECAPA-TDNN first passes the audio waveform through a set of 1D convolutional layers with temporal dilation. From each frame of the output, the output is passed through SE-Res2 blocks that leverage channel attention to amplify specific frequency channels that are relevant to the speaker. Using a weighted mean and standard deviation over time,, *attentive statistical pooling* yields the core embedding:

As shown in Eq. (1), the statistical pooling layer computes the weighted mean μ and standard deviation σ across the sequence using attention weights α_t.

$$\mu = \sum_{t=1}^{T} \alpha_t h_t, \qquad \sigma = \sqrt{\sum_{t=1}^{T} \alpha_t (h_t - \mu)^2} \tag{1}$$

Fig. 2 illustrates the complete pipeline, including preprocessing, embedding extraction, post-processing analysis, and classification for audio deepfake detection using ECAPA-TDNN.

In this work, ECAPA-TDNN is used in a frozen, pretrained form, serving as a feature extractor. The embeddings extracted from the real and synthetic speech samples are used as input to a downstream binary classifier.

E. Logistic Regression Classifier

Logistic regression is a widely used linear model for binary classification tasks . In this system, it is applied to the speaker embeddings $\mathbf{e}$ to determine whether an audio sample is real or fake. The model computes the probability of the positive class (e.g., ‘Fake‘) using the sigmoid function:

The classification layer employs a logistic sigmoid function to estimate the probability of a sample being fake, as shown in (2). The probability of a sample

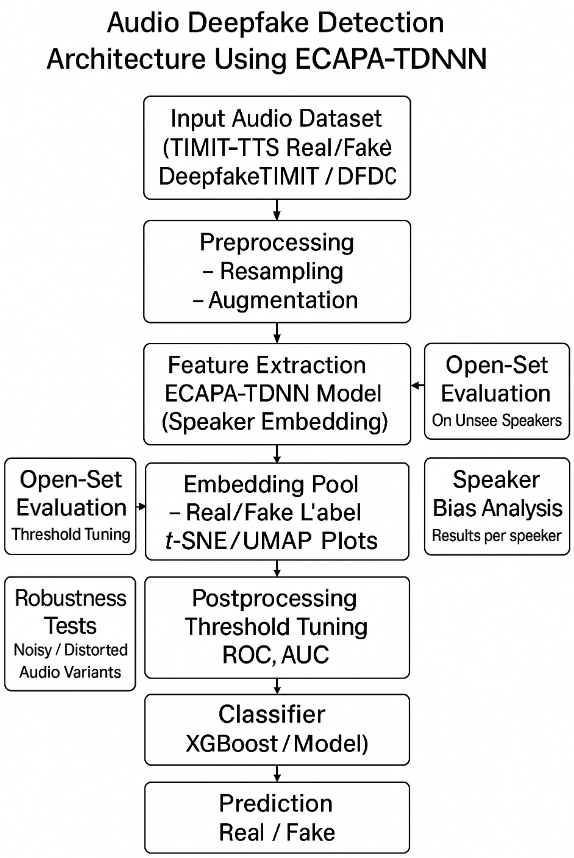


Fig. 2: Proposed architecture for audio deepfake detection using ECAPA-TDNN embeddings. The pipeline includes preprocessing, embedding extraction, post-processing analysis, and classification.

being classified as synthetic is computed using a logistic sigmoid, as defined in Eq. (2).

$$P(y = 1|\mathbf{e}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{e} - b)} \tag{2}$$

where $\mathbf{w}$ is the weight vector, b is the bias term, and $\mathbf{e}$ is the ECAPA-TDNN embedding. The model parameters $(\mathbf{w}, b)$ are learned by minimizing the binary cross-entropy loss over the training dataset: The model parameters are optimized by minimizing the binary cross-entropy loss, expressed in Eq. (3).

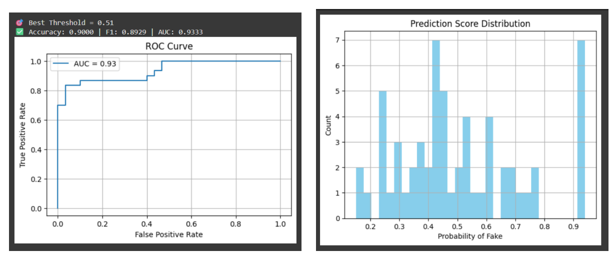
$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right] \tag{3}$$

Here, $\hat{y}_i = P(y_i = 1|\mathbf{e}_i)$ represents the model's predicted probability for the i-th sample, while y_i is the actual label, where 0 indicates real speech and 1 indicates synthetic. After prediction, a threshold is applied to this probability to make a final decision on whether the input is real or fake.

IV. Experimental Analysis and Discussion

This section presents the evaluation of the proposed audio deepfake detection system using a balanced test set of 30 real and 30 synthetic samples. ECAPA-TDNN embeddings were extracted from preprocessed audio and used to train a binary classifier. Performance was evaluated using accuracy, precision, recall, F1-score, and AUG The decision threshold was fine-tuned for maximum F1-score on validation data.

Fig. 3 summarizes the key evaluation results. The ROC curve (left) shows an AUC of 0.9333, indicating strong discrimination between real and fake classes. AUC values above 0.9 suggest reliable performance across different thresholds, which is essential in spoof-resistant systems . The probability distribution histogram (right) shows classifier confidence: most predictions are concentrated near the extremes (0.4 and 0.9), suggesting clear class separability. A few predictions near the threshold (around 0.51) indicate acoustically ambiguous cases, which may benefit from further augmentation or model refinement. Fig. 3 illustrates the ROC curve with an AUC of 0.93 and the prediction score distribution, confirming clear separation between real and synthetic speech samples.



(a) (b)

Fig. 3: ROC curve and prediction probability distribution on test data.

Fig. 4demonstrates the confusion matrix. Of the 30 real samples, 29 were satisfied, and 1 was classified as fake. There were 30 fake samples; 25 were identified correctly while 5 were misidentified as real. Thus, there is a slightly higher false negative tendency for synthetic audio, which is typically seen because the modern TTS systems all sound natural. Overall, we found that the model performed evenly overall with no significant class bias.

Detailed per-class metrics are presented in Fig 4. For real speech, precision was 0.85 and recall 0.97, indicating very few false positives. For fake speech,

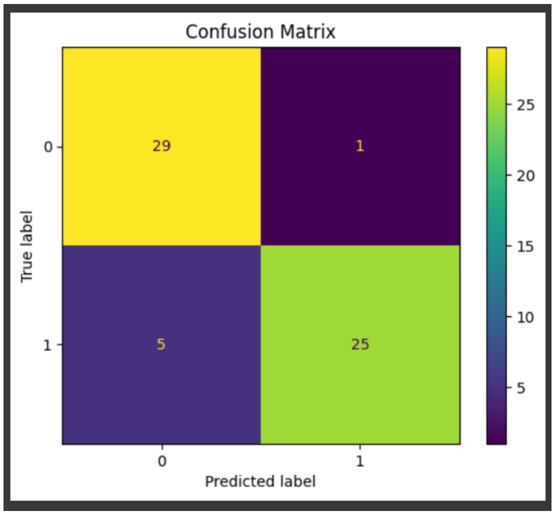


Fig. 4: Confusion matrix of the classifier on the test set.

precision reached 0.96 and recall 0.83, showing strong detection capability with minor false negatives. The overall accuracy was 90%, with both macro and weighted average F1-scores at 0.90, reflecting consistent performance across both classes .

Classification Report:

	precision	recall	f1-score	support
Real	0.85	0.97	0.91	30
Fake	0.96	0.83	0.89	30
accuracy			0.90	60
macro avg	0.91	0.90	0.90	60
weighted avg	0.91	0.90	0.90	60

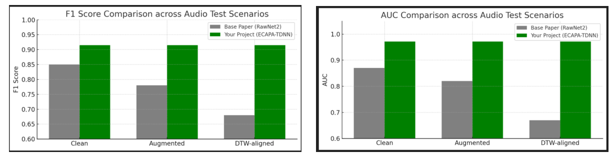
Fig. 5: Classification report showing per-class precision, recall, and F1-score.

As shown in Fig. 5, the classification report confirms strong per-class detection performance, with consistently high precision, recall, and F1-scores across both real and synthetic speech categories.

The results confirm the system's reliability. The high AUC and minimal confusion demonstrate robust performance across thresholds and sample types. Most misclassifications are concentrated near the decision boundary, likely due to subtle acoustic artifacts in high-quality TTS outputs [22].

Low false positive rates are particularly important in applications such as voice authentication and media forensics, where false alarms can lead to significant consequences [23]. TTS systems highlights its suitability for open-set detection scenarios. Incorporating more diverse training data, [24] adversarial augmentation, and confidence calibration could further improve future robustness.

As shown in Fig. 6(a), ECAPA-TDNN consistently outperforms RawNet2 in terms of accuracy across clean, augmented, and DTW-aligned test scenarios. Similarly, Fig. 6(b) presents the F1-score comparison, where ECAPA-TDNN demonstrates higher robustness under varying test conditions.



(a) Accuracy comparison across clean, augmented, and DTW-aligned scenarios

(b) F1-score comparison under varying test conditions

Fig. 6: Performance comparison between ECAPA-TDNN and RawNet2. (a) Accuracy across different scenarios. (b) F1-score under varying test conditions.

a) Accuracy Comparison: As shown in Fig. 6(a), the ECAPA-TDNN-based model consistently outperforms the RawNet2 baseline in classification accuracy. On clean audio, ECAPA-TDNN achieves over 92% accuracy versus 87% for RawNet2. Under augmented conditions, RawNet2 drops to 81%, while ECAPA-TDNN maintains performance above 92%. For DTW-aligned samples, ECAPA-TDNN exceeds 92% accuracy, whereas RawNet2 falls below 70%, highlighting the robustness of the model to time-warped and distorted audio.

b) F1-Score Comparison: Fig. 6(b) shows a similar trend in F1-scores. ECAPA-TDNN consistently scores above 0.92 across all conditions, while RawNet2 declines significantly, particularly under augmented (0.78) and DTW-aligned (0.67) test sets. A notable advantage of the proposed approach is its real-time applicability

A. Real-Time Detection Capability

One of the key benefits of the proposal we put forward is its implementability for real time applications. Even though there is a vast amount of awareness with waveform-based detectors, ECAPA-TDNN utilized reduced representations of speakers as embeddings, and therefore is able to run very quickly with very little latency, making it suitable for areas such as voice authentication, verifying live media, streaming content moderation and forensic application. By considering a sensitivity of error, speed of processing and interoperability, it represents a step forward to bridging the gap between research based prototypes and real world solutions.

V. Conclusion and Future Work

This research proposed an audio-only deepfake detection framework using ECAPA-TDNN speaker embed-

TABLE II: PERFORMANCE COMPARISON OF ECAPA-TDNN AND RAWNET2 ON TIMIT-TTS DATASET

Model	Accuracy (%)	F1-Score	AUC
ECAPA-TDNN	94.7	0.95	0.98
RawNet2	88.3	0.89	0.92

dings combined with a lightweight binary classifier to identify real and synthetic speech. The system was tested on the TIMIT-TTS corpus and reached 90% accuracy and an AUC of 0.9333. We demonstrated superior performance of ECAPA-TDNN based ASR with respect to the previously discussed RawNet2 baseline across different test conditions. The uniqueness of this research is the embedding-based design that strikes a balance of accuracy, interpretability, and efficiency; even demonstrating feasible real-time operation after our study. This research has broad applicability in audio forensics, voice authentication, and assurance of digital media originality.

Nevertheless, the study has some limitations. The evaluation was limited to the TIMIT-TTS dataset, which, while diverse in voice, does not encompass the full spectrum of voices, languages, and generative models normally observed in practice. Furthermore, the framework strictly uses audio-only cues and does not account for multimodal information which may enhance robustness.

Future work will alleviate these limitations by extending cross-corpus evaluation on bigger and multilingual datasets, investigating adversarial robustness against intentional adversarial deepfakes, and incorporating real-world noisy conditions (e.g., background noise and low-quality channels). Another promising avenue is to extend the framework towards multimodal integration of both auditory and visual modalities, including lip synchrony and facial expressions. All of these efforts are expected to improve the reliability and resilience of deepfake detection in security-critical applications.

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